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MyDesign Portfolio

Element E Initial Template

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

STEM Principle for High-Priority Design Requirement #1

STEM Concept:

Measuring

Describe how STEM concept applies directly to the specifics of your design:

Measuring is a STEM concept that directly applies to the specifics of our design because we need to be able to take accurate measurements to model the design and complete the building process.

STEM Principle for High-Priority Design Requirement #2

STEM Concept:

Testing the prototype to see if it works

Describe how STEM concept applies directly to the specifics of your design:

The STEM concept of testing a design prototype applies directly to the specifics of our design because we need to test our design prior to finalizing it to make sure it meets the design requirements.

STEM Principle for High-Priority Design Requirement #3

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STEM Concept:

Research

The STEM concept of researching applies directly to the specifics of our design because we need to be able to conduct research in order to build an effective design.

Proposed Solution to STEM Principles and Practices (E2)*STEM Principle for Design Solution #1*

STEM Concept:

Measuring

Describe how STEM concept applies directly to the specifics of your solution:

Without measurements we wouldn't be able to make a design that can accurately meet the design requirements.

STEM Principle for Design Solution #2 - Iteration of the first design

STEM Concept:

Testing the prototype to see if it works

Describe how STEM concept applies directly to the specifics of your solution:

Without the ability to test our design we couldn't fix the flaws that may show up during a test, emphasizing how testing the design is vital.

STEM Principle for Design Solution #3 - Iteration of the second design

STEM Concept:

Research

Describe how STEM concept applies directly to the specifics of your solution:

Research will greatly help us in figuring out how to build or fix certain issues with our design.

Evidence Design is Reviewed (E3)

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The expert confirmed that our design requirements align with STEM principles, particularly in fluid dynamics, material science, and automation. They emphasized the importance of selecting mold-resistant materials and ensuring efficient water distribution to prevent excess moisture accumulation.

Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principle for Design Solution #1

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Fluid dynamics and material science.

Describe how STEM concept applies directly to the specifics of your solution:

Our initial design applies fluid dynamics by ensuring even water distribution to prevent excess moisture, reducing mold risk. We also use hydrophobic materials to resist mold



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growth.

STEM Principle for Design Solution #2 - Iteration of the first design

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Sensor-based automation and environmental control.

Describe how STEM concept applies directly to the specifics of your solution:

In the second iteration, we integrated sensors to automate water release, maintaining optimal moisture levels and preventing overwatering, which can lead to mold formation.

STEM Principle for Design Solution #3 - Iteration of the second design

STEM Concept

UV filtration and chemical treatment integration.

Describe how STEM concept applies directly to the specifics of your solution:

The third iteration includes UV filtration and mild chemical treatments to further inhibit mold growth while ensuring safe water for plant use.



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Evidence Design is Reviewed (E3)*Documentation #1:*

- What format:

Interview notes

The expert confirmed that the selection of mold-resistant materials aligns with engineering principles in material science, emphasizing the importance of non-porous surfaces and antimicrobial coatings to prevent mold growth.

Documentation #2:

- What format: Online Interview

The expert stated that even water distribution relies on agricultural engineering principles, particularly irrigation technology used in commercial greenhouses. They noted that drip irrigation or misting systems could enhance efficiency while preventing overwatering.

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- The expert mentioned that designing for plant adaptability integrates botanical research, where varying water needs of plants should be met through adjustable mechanisms, such as zone-controlled irrigation.



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Documentation #3:

- What format:

Video call transcript

The expert acknowledged that durability and longevity factors depend on material science and environmental sustainability. They recommended UV-resistant and corrosion-proof materials to extend the system's lifespan.

Experts with Credentials (E4)

Expert #1: A mechanical engineer specializing in fluid dynamics and irrigation system design. Has studied Mechanical Engineering with a focus on sustainable water management.

Expert #2: An agricultural scientist with expertise in plant hydration needs and greenhouse management. Has experience in commercial irrigation systems.

Expert #3: A materials scientist with a background in polymer chemistry and antimicrobial coatings. Works in developing water-resistant materials for industrial applications.

Verification or Corrective Response to Design Review (E5)

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Interview with Expert #1 informs corrective response: Based on the recommendation regarding fluid dynamics, the team will explore the integration of a pressure-regulated watering system to improve efficiency and prevent stagnant water.

Interview with Expert #2 informs corrective response: Given the insights on agricultural engineering, the design will incorporate a drip irrigation mechanism for more precise water control and plant-specific hydration.

Interview with Expert #3 informs corrective response: After learning about material science considerations, the system will now include UV-resistant, corrosion-proof materials to enhance durability and long-term functionality.